

Electronic, Electrical and Systems Engineering
Risk assessment for generically assessed operations

- ✓ Using equipment that has been purchased and provided by the School
- ✓ Using equipment that has a valid tested for electrical safety label
- ✓ Using a bench power supply that provides less than 30 Volts between terminals or earth
- ✓ Working on a project that USES voltages below 30 Volts (whether generated or transformed).
- ✓ Working on a project that produces/uses Radio frequency signals below 1mW/cm²
- ✓ Using a P.C. solely to write software/documentation
- ✓ Using lasers classified as Class 1
- ✓ Only Soldering in a designated area, with fume extraction & using colophony free solder
- ✓ Only using small hand-tools to construct electronic circuits (Pliers, wire cutters, and screwdrivers)
- ✓ Working with items at temperatures between 5 degrees and 30 degrees Celsius
- ✗ Not working with revolving or moving parts
- ✗ Not using Chemicals (Including proprietary substances such as glue, cleaning solvents or water)
- ✗ No Eating, Drinking and open toe sandals

Any work outside these parameters must not be carried out until a risk assessment has been made.

In some cases, the risk assessment may be simple clarification from your supervisor or the person responsible for the working area, for example, does a piece of battery operated equipment require testing for electrical safety.

In ALL other cases a full risk assessment will need to be carried out detailing the Hazard and likelihood of the risk, this may need to take into account legislation such as the Electricity at work regulation, The COSHH (Control of substances Hazardous to Health Regulations etc) and the School / University Health and Safety policies.

By signing this form, you are

- a) Stating that all of your work confirms to the above parameters
- b) Stating that you have read the relevant School and University Health and Safety Policies.

This form must be signed, or a written Risk assessment carried out and presented to your supervisor before you start any work and a new risk assessment made whenever circumstances change

Access into swipe operated laboratories will not be allowed until a written risk assessment is produced.

The Student's Project Laboratory 125 is generally designed for the use of UG and MSc who have signed this form and do not require a written risk assessment. However, some activities outside these parameters may be allowed following a risk assessment.

UG & MSc students operating outside these parameters must work in a designated research area and must display a copy of their risk assessment prominently at their working position.

I agree that I have assessed my work, and it only involves those operation stated above

I understand that should my work change such that I am operating outside the above parameters I will need to re-asses the risks.

Location: Y3-LF03

Supervisor Name: Dr. Yongjing Wang

Student Name: Chissanupong Saengsint

Supervisor Signature: Yongjing Wang

Student ID Card: 2881058

Date: 14/11/2025

Student Signature: Chissanupong

Date: 13/11/2025

THE UNIVERSITY OF BIRMINGHAM

SCHOOL OF ENGINEERING – MSc.Dissertation Projects

Full Risk Assessment

Student Name and ID: Chissanupong Saengsint 2881058

The Risk Assessment number will be allocated by the local H&S Coordinator

The RA must be signed by the PI or Supervisor to approve it being suitable and sufficient.

Each person named below must have their own copy of the RA, any subsequent changes or reviews must received by each person named below.

Persons covered by this assessment

Name	Signature	Date
Chissanupong Saengsint		13/11/2025
Dr.Yongjing Wang		14/11/2025

All personnel will refuse to carry out procedures they deem unsafe or do not feel suitably trained to carry out.

All University of Birmingham staff and visitors are covered under the University of Birmingham's insurance policy.

All personnel should make themselves aware of the School of Engineering's Health and Safety Code of Practice

Brief Description of project

The aim of this project is to study robotic skill transfer using kinematic learning between different robotic platforms. The focus is to evaluate whether a model trained on a 7 DOF robotic arm can be effectively adapted to other robots, specifically the KUKA KR R14, with reduced training effort.

Method Statement

The experiment involves using the KUKA KR R14 robot in the UKRRIN laboratory to perform reach tasks, where the end effector moves to specific target positions in space. These movements are generated from a kinematic control model developed in simulation and applied on the physical robot to verify transfer accuracy.

The project does not involve physical manipulation, tools, or contact operations. Only motion testing within the robot's workspace will be carried out. All activities take place inside the enclosed robotic workstation under standard laboratory safety protocols.

Key activities of the robot arm inside the workstation include:

- 1. Moving the robot end effector to predefined 3D positions based on the trained kinematic model.*
- 2. Recording joint configurations, positional accuracy, and motion smoothness for each target position.*
- 3. Repeating the same motions using reference and transferred models for performance comparison.*

Key activities carried out by the operator outside the workstation include:

- 1. Uploading and adjusting robot programs on the PC.*
- 2. Running simulations using PC on the ISACC Lab and ROS2 workspace to generate kinematic trajectories.*
- 3. Monitoring robot movements through the workstation window and ensuring safety interlocks remain active.*
- 4. Collecting and analyzing positional data for evaluating model transfer accuracy.*

GENERAL HEALTH AND SAFETY RISK ASSESSMENT FORM

Site: School of engineering Y3

Department: School of Engineering

Activity: LF03, using robot arm to conduct experiment

Supervisors Name: Dr. Yonjing Wang

Hazard Assessment				Control Assessment									Actions		
Hazard Category	Hazards Identified	Who might be harmed? Staff Students Contractors Others	How might people be harmed?	Existing Control Measures	Initial Risk Rating			Are these adequate? Yes/No	Changes to/ Additional Controls	Residual Risk Rating			Owner	Due Date	Action Complete
					S	L	R			S	L	R			
Physical hazards	Noise exposure	Staff and students	Hearing impairment	The workstation is fully enclosed to reduce noise levels at the source.	1	2	2	yes							
Physical hazards	Collision from robots in motion Collision between the robotic arm or its components, such as the frame and objects. Collision between the tools attached Collision between the tools attached and the tool changer.	Staff and students	Cause unexpected malfunctions or create hazardous situations in the robot's environment, leading to potential harm to people in the vicinity.	Programming and motion planning: Ensure that the robot's programming and motion planning considers the workspace limitations, object detection, and collision avoidance to prevent accidental collisions. Or trigger an immediate stop to prevent damage or injury. Workers stay a safe distance to the robot (stand outside the barriers) in testing. Safety button is installed on the cell.	2	2	4	yes							
Physical hazards	Falling objects from height –	Staff and students	Cause impact injuries, such as bruises, cuts, or fractures, if it strikes a person's foot with force.	Covered shoes must always be worn to protect feet.	1	2	2	yes							

Hazard Assessment				Control Assessment									Actions		
Hazard Category	Hazards Identified	Who might be harmed? Staff Students Contractors Others	How might people be harmed?	Existing Control Measures	Initial Risk Rating			Are these adequate? Yes/No	Changes to/ Additional Controls	Residual Risk Rating			Owner	Due Date	Action Complete
					S	L	R			S	L	R			
Physical hazards	Electrical tools - Entanglement/s hock from torque tools	Staff and students	Create potential electrical hazards that can harm people through shocks or fires	Tools are CE marked, fused, and insulated. Knowledge of the tool is required before operating.	1	2	2	yes							
Physical hazards	Electrical shock from robots, electrical tools, controllers, PC, and all equipment connected to an energy source	Staff and students	Create potential electrical hazards that can harm people through shocks or fires	Key equipment is CE-marked, fused, insulated and PAT-tested. The workstation is to be disconnected from the mains supply before any configuration work is undertaken and when left unattended. 230V isolation switch is mounted on the wall near the cell. No lone working allowed.	3	2	6	yes							
Ergonomi c hazards	Risk of ergonomic injuries from lifting and carrying batteries dummy, tools and components for building the frame/ robotic cell.	Staff and students	Ergonomic injuries	The tools used on robot should be less than 3 Kgs,	1	1	2	yes							
Physical hazards	Trap and pinch when moving components.	Staff and students	Lead to pinching or entanglement of clothing, or body parts, which can cause injuries	Maintain a distance of at least 0.5 m during the operation of the workstation. For demonstrations, the enclosure should be used. Workstation is needed to be un-powered when adjusting the mechanical configuration.	2	2	4	yes							
Physical hazards	Slips, trips, and falls - Trip over comms/power	Staff and students	Slips, trips, and falls	Good housekeeping. The wires and cable are tied back properly. A cable protection ramp is used to protect cables and reduce trip risk.	2	2	4	yes							

Hazard Assessment				Control Assessment								Actions			
Hazard Category	Hazards Identified	Who might be harmed? Staff Students Contractors Others	How might people be harmed?	Existing Control Measures	Initial Risk Rating			Are these adequate? Yes/No	Changes to/ Additional Controls	Residual Risk Rating			Owner	Due Date	Action Complete
					S	L	R			S	L	R			
	cables on floor working area														

	Consequence / Severity score (severity levels) and examples of descriptors				
	1	2	3	4	5
Domains	Negligible	Minor	Moderate	Major	Catastrophic
Impact on the safety of staff, students or public (physical / psychological harm)	Minimal injury not requiring first aid or requiring no/minimal intervention or treatment. No time off work	Minor injury or illness, first aid treatment needed or requiring minor intervention. Requiring time off work for <3 days	Moderate injury requiring professional intervention Requiring time off work for 4-14 days RIDDOR / MHRA / agency reportable incident	Major injury leading to long-term incapacity/ disability (loss of limb) Requiring time off work for >14 days	Incident leading to death Multiple permanent injuries or irreversible health effects
E.g. for a Dissertation success	None	Significant late night working to resolve	Insufficient data gathered Poor quality submission / Rushed Submission	Failure of Dissertation / Reputation Damage to people / companies. Late submission	Actual personal danger to those involved (e.g. politically dangerous questions that may result in repercussion)

Likelihood score	1	2	3	4	5
Frequency	Rare	Unlikely	Possible	Likely	Almost certain
Broad descriptor	This will probably never happen/occur	Do not expect it to happen/occur but it is possible it may do so	Might happen or occur occasionally	Will probably happen/occur but it is not a persisting issue	Will undoubtedly happen/occur, possibly frequently
Time-framed descriptor	Not expected to occur for years	Expected to occur at least annually	Expected to occur at least monthly	Expected to occur at least weekly	Expected to occur at least daily
Probability Will it happen or not?	<0.1 per cent	0.1–1 per cent	1.1–10 per cent	11–50 per cent	>50 per cent

The overall **level of risk** is then calculated by multiplying the two scores together.

Risk Level = Consequence / Severity x Likelihood (R = C x L)

		Likelihood (L)					REGION
Likelihood score		1	2	3	4	5	
Description		Rare	Unlikely	Possible	Likely	Almost certain	
Consequence (C)							
5 Catastrophic		5	10	15	20	25	Intolerable – Cease activity consult with supervisor directly Intolerable – significant change to activities required – consult with supervisor Tolerable – attempt reasonable improvement Negligible – no action really necessary, but improvement may be attempted.
4 Major		4	8	12	16	20	
3 Moderate		3	6	9	12	15	
2 Minor		2	4	6	8	10	
1 Negligible		1	2	3	4	5	